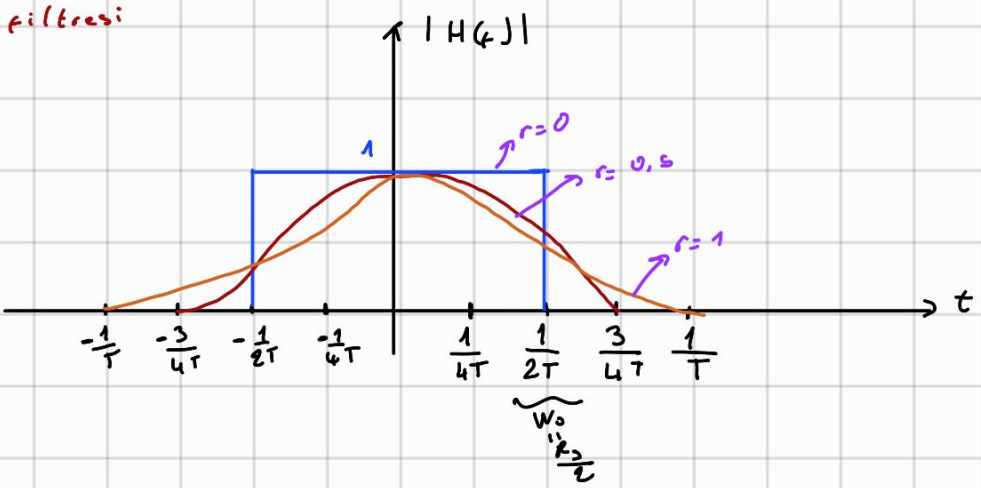
Raised Cosine (RC) Pulse

Yükseltilmiş kosinüs darbesi

$$H(f) = \begin{cases} 1 & , |f| < 2W_0 - W \\ \cos^2\left(\frac{\pi}{4}\,\frac{|f|+W-2W_0}{W-W_0}\right) & , 2W_0 - W < |f| < |W| \\ 0 & , |f| > W \end{cases}$$

RC filtresi



$W_0 = \frac{R_s}{2} = \frac{1}{2T}$ min Nyquist BW for rectangular spectrum

$W - W_0$ = Aşırı BG (Excess BW)

$r = \frac{W - W_0}{W_0}$ Azalma faktörü (roll-off factor)

$0 \le r \le 1$

$W = W_0(1+r)$

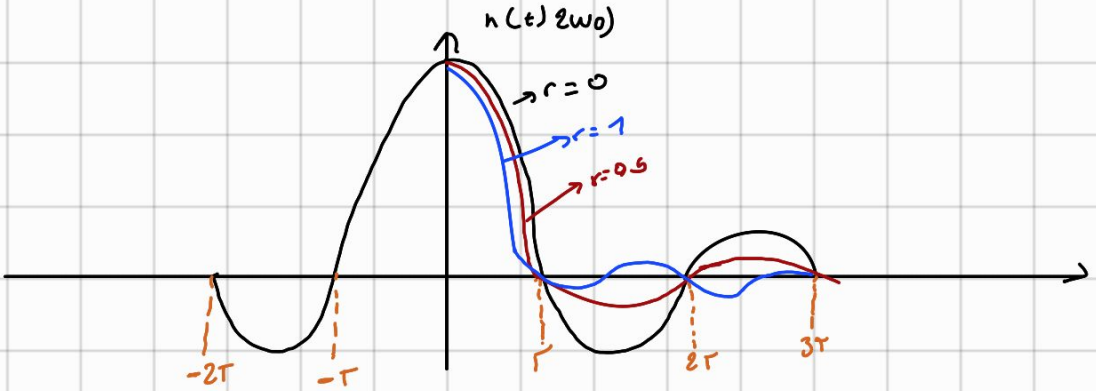
$$\boxed{W = \frac{R_s}{2}(1+r)\ Hz}$$ → Temel band haberleşmesinde RC darbesinin BG

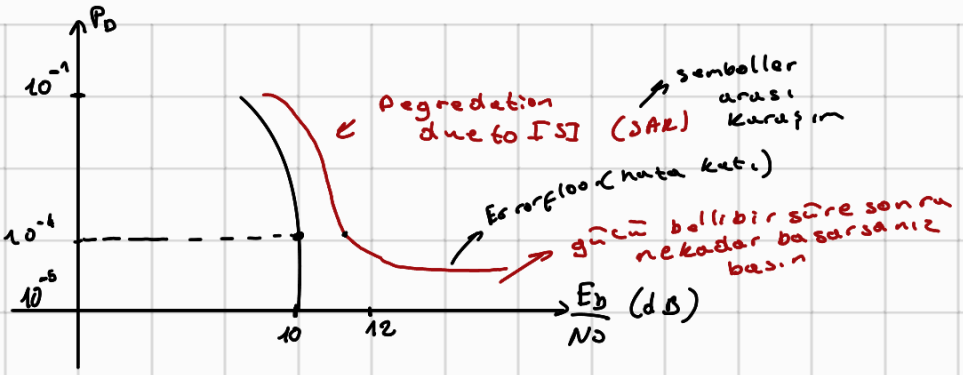
Mutlak BG (Absolute BW)

→ Band geçiren Haberleşme

$W_{GYB} = R_s(1+r)\ Hz$ // (Taşıyıcılı haberleşme)

$h(t) = 2W_0\,\mathrm{sinc}(2W_0 t)\,\frac{\cos[2\pi(W-W_0)t]}{1-[4(W-W_0)t]^2}$ azalma hızı $\frac{1}{t^3}$





güç → equalization

$$H(f) = H_t(f)\, H_c(f)\, H_r(f) \qquad \Rightarrow H(f) = H_t(f)\, H_r'(f)$$

H_c: channel; $H_r(f) = H_c(f)\, H_r'(f)$, $H_c(f) = \frac{1}{H_c(f)}$

$H(f)$: RC; $H_t(f)$: Root RC (RRC); $H_r'(f)$: RRC

$h_r'(t) = h_t(T-t)$ karşılaştırıcı filtre

Önce S143 Ex 3.3

Example 3.3 Bandwidth Requirements

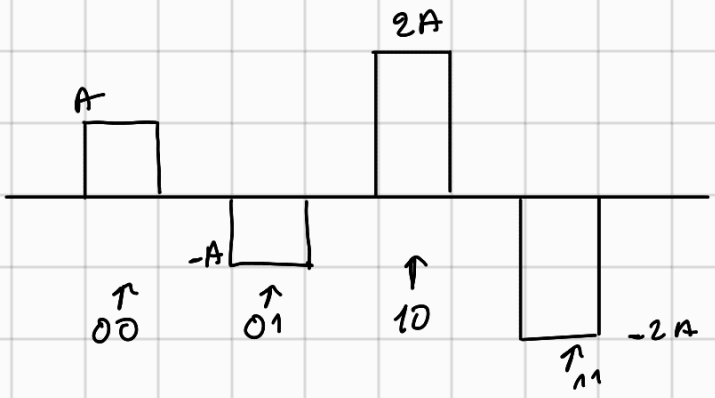
(a) Find the minimum required bandwidth for the baseband transmission of a four-level PAM pulse sequence having a data rate of $R = 2400$ bits/s if the system transfer characteristic consists of a raised-cosine spectrum with 100% excess bandwidth ($r = 1$).

(b) The same 4-ary PAM sequence is modulated onto a carrier wave, so that the baseband spectrum is shifted and centered at frequency f_0. Find the minimum required DSB bandwidth for transmitting the modulated PAM sequence. Assume that the system transfer characteristic is the same as in part (a).

4 level PAM

$R = 2400$ bits/s

RC 100% excess BW ($r = 1$)



$M = 2^k = 4 \Rightarrow k = 2$ bits/symbol (pulse)

$$R_s = \frac{R_b \text{ bit/sn}}{k \text{ bits/sembol}} = \frac{2400 \text{ bit/sn}}{2 \text{ bit/sn}} = 1200 \frac{\text{sembol}}{\text{sn}} \text{ (darbe)}$$

R_s: sembol hızı

$$W = \frac{R_s}{2}(1+r) = \frac{1200}{2}(1+1) = 1200 \text{ Hz}$$

b) $W_{DSB} = R_s(1+r) = 2400$ Hz

Kontrollü SAK ile işaretleşme

(Signaling with controlled ISI)

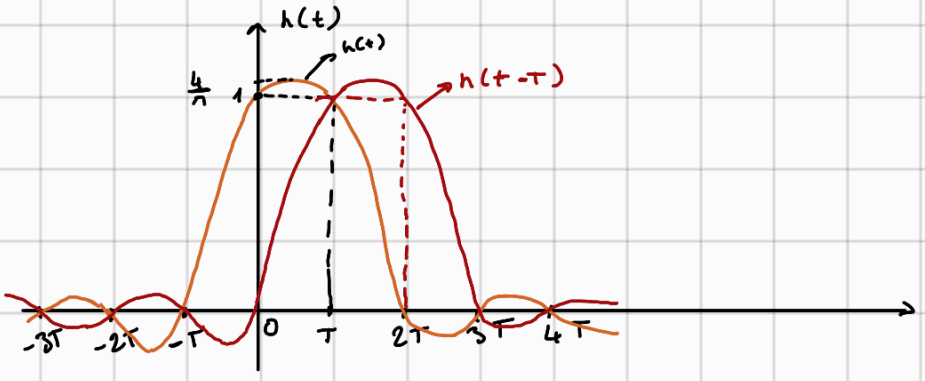
(kısmi cevaplı işaretleşme - Partial response signaling)

Sıfır SAK için 2. Nyquist kriteri

$$h(nT) = \begin{cases} 1 & , n = 0, 1 \\ 0 & , \text{Diğer} \end{cases}$$

Çift kutuplu

"1" → h(t)
"0" → -h(t)



Herikî darbbe + ⇒ Örnekle

" " " - ⇒ " = -2

Darbeler zıt işaretli ⇒ " = 0

Eğer örnek > 0 ⇒ Mevcut bit ve önceki bit "1"

örnek < 0 ⇒ " " " " " "0"

" = 0 ⇒ Mevcut bit önceki bitin tümleyenidir

İlletilendizi :	1	1	0	1	1	0	0	0	1	0	1	1	1
Örnekler :	1	2	0	0	2	0	-2	-2	0	0	0	2	2
Tespit edilen dizi :	1	1	0	1	1	0	0	0	1	0	1	1	1

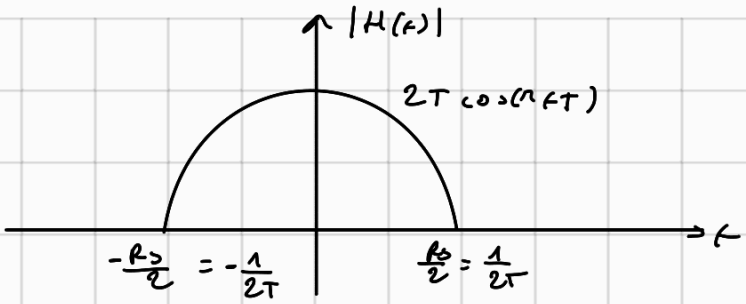
Duobinary darbe

$$h(t) = \frac{\sin(\pi R_s t)}{\pi R_s t (1 - R_s t)} = \frac{\sin(\pi t/T)}{\frac{\pi t}{T}\left(1 - \frac{t}{T}\right)} = \underbrace{\text{sinc}\left(\frac{t}{T}\right) + \text{sinc}\left(\frac{t-T}{T}\right)}_{\text{sklar, s. 97}}$$

↳ Azalma hızı $\frac{1}{T^2}$

$$h(t) \xrightarrow{F} H(f) = \frac{2}{R_s}\cos\left(\frac{\pi f}{R_s}\right)\Pi\left(\frac{f}{R_s}\right)e^{-j\frac{\pi f}{R_s}}$$

$$= 2T\cos(\pi fT)\,\Pi(fT)\,e^{-j\pi fT}$$



kosinüs filtresi

RC darbesinden daha az BG'ye sahip

bu filtre $\frac{1}{t^2}$ ile azalmakta senkronizasyon hataları

daha fazla RC'de $\frac{1}{t^3}$ ile azalmakta

S 164 Pr 3.8

3.8. **(a)** What is the theoretical minimum system bandwidth needed for a 10-Mbits/s signal using 16-level PAM without ISI?
(b) How large can the filter roll-off factor be if the allowable system bandwidth is 1.375 MHz?

a) $R = 10$ Mbit/s

16 level PAM w/o ISI

$M = 2^k = 16 \rightarrow k = 4$ bits/symbol (pulse)

$R_s = \frac{R}{k} = \frac{10 \text{ Mbits/s}}{4 \text{ bits/symbol}} = 2,5 \text{ M} \frac{\text{symbols}}{\text{s}}$

$W = \frac{R_s}{2} = 1,25$ MHz

b) $W = 1,375$ MHz

$W = \frac{R_s}{2}(1+r) \rightarrow r = 0,1$

$\frac{1,375}{1,25}$

Pr. 3.9

Q dist $\leq \pm 0,1\%$ of peak to peak signal volage

$f = 8000$ samples/s

$M = 32$ PAM

3.9. A voice signal (300 to 3300 Hz) is digitized such that the quantization distortion $\leq \pm 0.1\%$ of the peak-to-peak signal voltage. Assume a sampling rate of 8000 samples/s and a multilevel PAM waveform with $M = 32$ levels. Find the theoretical minimum system bandwidth that avoids ISI.

Min theo BW w/o ISI

$\Delta v = \frac{2 \times p}{L} \rightarrow \frac{\Delta v}{2} = \frac{\times p}{L} \leq \frac{0,1}{100} 2 \times p \rightarrow L \geq \frac{100}{0,2} = 500$

$\underbrace{2^n}_{512} \geq L \geq 500 \Rightarrow 2^n = 512 \Rightarrow n = 9$ bits/sample

$R_{bit} = 9 \frac{\text{bits}}{\text{sample}} \times 8000 \frac{\text{samples}}{\text{s}} = 72$ k bits/s

$M = 32 = 2^5 \rightarrow k = 5$ bits/symbol (pulse)

BG = 36 kHz

Skat BG düştü

$$R_s = \frac{R_b}{k} = \frac{72k\ bits/s}{5\ bits/symbol} = 14\,400\ \frac{symbols}{s} \rightarrow W = \frac{R_s}{2} = 7200\ Hz$$

↑ Bit hata olasılığı daha kötü

(1)0

Örnek

In a CD recording system, 2 voice signals are sampled with a sampling freq. of 44,1 kHz, multiplexed and encoded with 16 bits.

a) What is the bit rate?

b) How many bits are needed for a recording of 50 minutes?

a) $R_b = 2 \cdot n \cdot f_s = 2 \cdot 16\ \frac{bits}{sample} \cdot 44{,}1\ \frac{ksample}{sec} = 1\,411\,200\ \frac{bits}{s}$ (2 = signal sayısı)

b) # of bits $= 1\,411\,200\ \frac{bits}{s} \cdot 50 \times 60\ s = 4\,233\,600\,000$ bits

$\frac{}{8}$'e bölelim = 529,2 M byte

sonra 1000'e

3.9. A voice signal (300 to 3300 Hz) is digitized such that the quantization distortion $\leq \pm 0.1\%$ of the peak-to-peak signal voltage. Assume a sampling rate of 8000 samples/s and a multilevel PAM waveform with $M = 32$ levels. Find the theoretical minimum system bandwidth that avoids ISI.

Min theo BW w/o ISI

$$\Delta v = \frac{2 \times p}{L} \rightarrow \frac{\Delta v}{2} = \frac{xp}{L} \leq \frac{0{,}1}{100} 2 \times p \rightarrow L \geq \frac{100}{\Delta 2} = 500$$

$$\underbrace{2^n}_{512} \geq L \geq 500 \quad \Rightarrow 2^n = 512 \Rightarrow n = 9\ bits/sample$$

$$R_b = n f_s = 9\ \frac{bits}{sample} \times 8000\ \frac{samples}{s} = 72\ k\ bits/s \qquad BG = 36\ kHz$$

$M = 32 = 2^5 \Rightarrow k = 5\ bits/symbol\ (pulse)$

5 kat BG düştü

$$R_s = \frac{R_b}{k} = \frac{72k\ bits/s}{5\ bits/symbol} = 14\,400\ \frac{symbols}{s} \rightarrow W = \frac{R_s}{2} = 7200\ Hz$$

↑ Bit hata olasılığı daha kötü

c) $r = 0{,}25 \Rightarrow$ Bipolar signaling BG?

$R_b = 72k\ bits/s \rightarrow W = \frac{R_b}{2}(1+r) = \frac{72k}{2}(1+0{,}25) = 45\ kHz$

$r = 0{,}25 \rightarrow M = 32$ PAM BG ? $R_s = 14\,400$ symbols/s $\rightarrow W = \frac{R_s}{2}(1+r) = \frac{14\,400}{2}(1+0{,}25)$

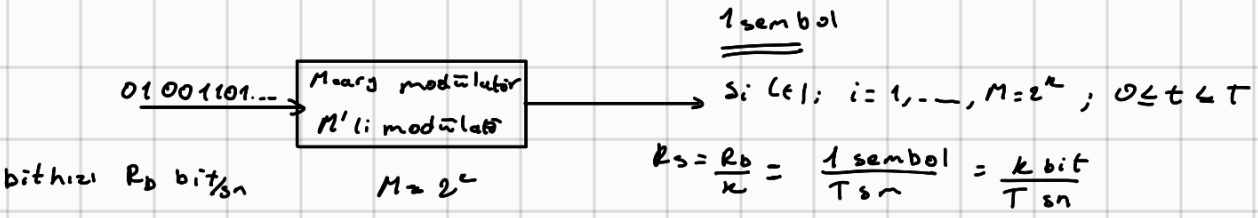
$= 9\ kHz$ //

Sayısal Taşıyıcılı Modülasyon

(Carrier Mod / Bandpass Mod)

Faz Kaydırmalı Anahtarlama (FKA) Mod

Phase Shift keying (PSK) Mod



bithızı R_b bit/sn $\quad M = 2^k$

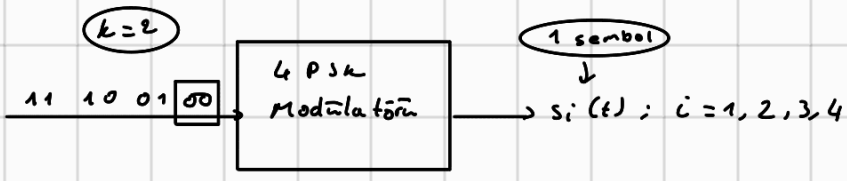
$R_s = \frac{R_b}{k} = \frac{1 \text{ sembol}}{T \text{ sn}} = \frac{k \text{ bit}}{T \text{ sn}}$

PSK: $s_i(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \phi_i) \quad ; \quad 0 \le t \le T$

$i = 1, \dots, M = 2^k$

$\phi_i = \frac{2\pi(i-1)}{M}$

Örnek $M = 4 = 2^k \Rightarrow k = 2$ bit/sembol



$00 \rightarrow s_1(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + 0)$; $0 \le t \le T$

$01 \rightarrow s_2(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \frac{\pi}{2})$; $0 \le t \le T$

$10 \rightarrow s_3(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \pi)$; //

$11 \rightarrow s_4(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \frac{3\pi}{2})$; //

Enerji

$E_{si} = \underbrace{\frac{2E}{T} \frac{1}{2}}_{\frac{A^2}{2}} \cdot T = E$

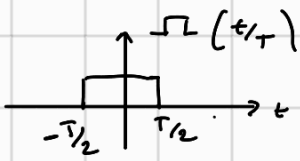
$\forall \; E_{si} = E$

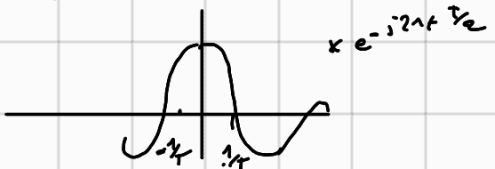
$E_{si} = \int_0^T s_i^2(t)\,dt = \frac{2E}{T} \int_0^T \underbrace{\cos^2(2\pi f_c t + \phi_i)}_{\frac{1}{2}[1 + \cos(4\pi f_c t)]} dt = \frac{2E}{T} \frac{1}{2} \int_0^T dt = E$

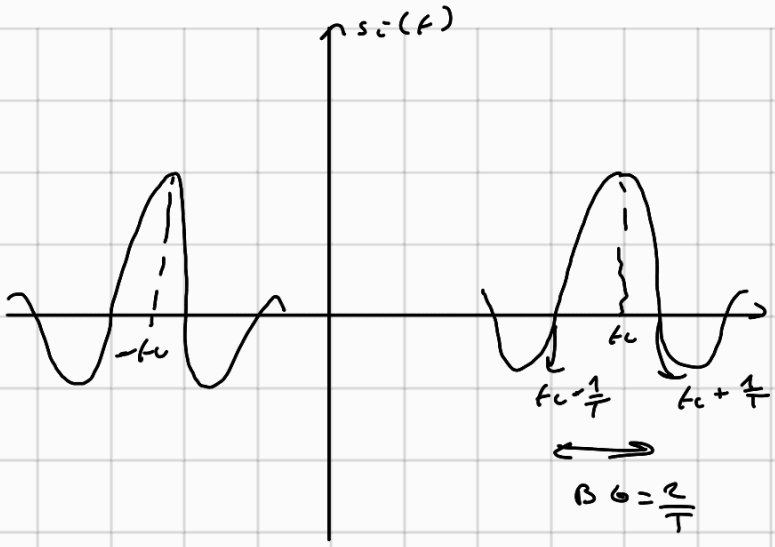
$\int_0^T \cos(4\pi f_c t)\,dt \approx 0$

BG

$s_i(t) = \sqrt{\frac{2E}{T}} \; \Pi\left(\frac{t - T/2}{T}\right) \cos(2\pi f_c t + \phi_i)$







M arttığında BG aynı kalıyor bu önemli bir avantaj

Bant genişliği verimli teknik

$$BGV = \frac{R_b \text{ bit/sn}}{W \text{ Hz}} \quad \frac{\text{bit/sn}}{\text{Hz}}$$

→ bir Hz bant genişliğinde bir saniyede gönderilen bit

BGV verimliliği: $\frac{1}{BG}$

MPSK: M↑ ⇒ BG aynı → BGV ↑ BGV verimli teknik (BW efficient mod technique) × P_b ↑ performans kötüleşiyor

Bant genişliğinden kazanıyorum güçten kaybediyorum

Özel durum

k=1, M=2 ⇒ İkili FKA = iFKA

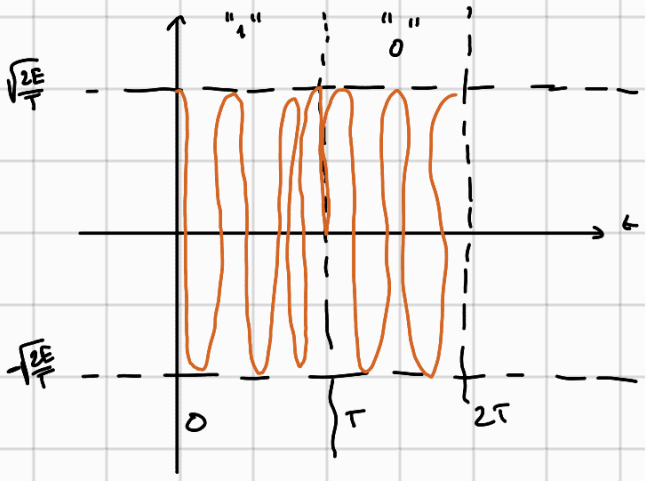
Binary PSK = BPSK

"1" → $s_1(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + 0)$; $0 \le t \le T$

"0" → $s_2(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \pi)$; $0 \le t \le T$

$= -\sqrt{\frac{2E}{T}} \cos(2\pi f_c t)$; $0 \le t \le T$

Bipolar signaling x carrier → $P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$ $E_b = E$



→ s174 şekil 4.5

Ortogonal modülasyon

Frekans kaydırmalı Anahtarlama (FRKA)

Frequency Shift Keying (FSK)

$$s_i(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_i t), \quad 0 \le t \le T, \quad i = 1,2,3,4 \dots$$

$$\hookrightarrow f_i = f_c + (i-1)\Delta f$$

$$\int_0^T s_i(t) \cdot s_j(t)\,dt = \begin{cases} 0 & i \neq j \\ E & i = j \end{cases} \Rightarrow \text{ortogonal (dik)}$$

Ortogonallik şartı:

$$\Delta f = \begin{cases} \frac{1}{2T} & \text{Faz tahminli demod (Coherent demod)} \quad BG/2 \\ \frac{1}{T} & \text{" tahminsiz " (Non coherent demod)} \quad BG \end{cases}$$

→ Sayfa 202 Ex 4.3

$M \uparrow \Rightarrow BG \uparrow \Rightarrow BGV \downarrow \quad \times P_b \downarrow$

(Power efficient Mod.)